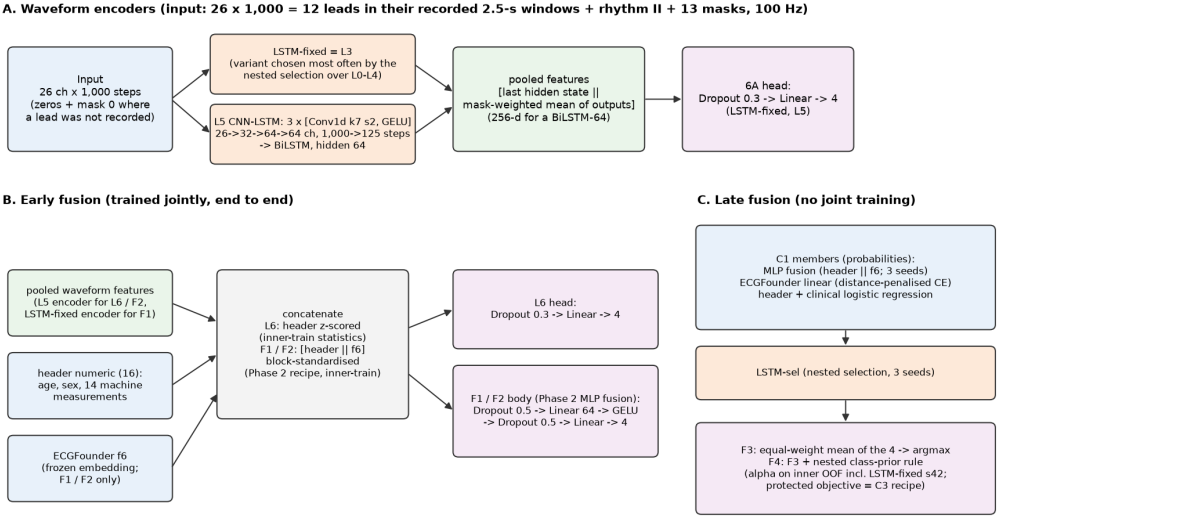


Training of every network: class-balanced cross-entropy, AdamW (lr 1e-3, weight decay 1e-2), batch 32, gradient clipping 1.0, <= 60 epochs, early stopping on the fold's inner-validation S3 (patience 12, min 10 epochs), best epoch restored; seeds 42/43/44 x CV repeats 0-2 (patient-grouped 5-fold); all standardisation statistics from the inner-training rows only; the outer test fold is predicted once. Development set only (locked test never used).